Polymeric Chemistry
DSC and TGA

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Abstract

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1 Results

Two different samples were prepared, and analyzed, using DSC and TGA. The Data of the samples (PET and *it*-PS) are listed in the Tables 1 and 2 below.

Tab. 1: Listed are the meltingtemperatures(T_m), glastemperatures(T_g) and the melting enthalpy (Δ H$_m$) of the heatingcurves if *it*-PS and PET.

Sample	Δ H$_m$ [$\frac{J}{g}$]	T$_m$ [°C]	T$_g$ [°C]
PET 1.hc	38.368	249.85	-
PET 2.hc	32.360	246.00	84.36
it-PS 1.hc	24.194	221.61	99.69

Tab. 2: Listed is the crystallizationtemperature(T_c) and the crystallization enthalpy (Δ H$_c$) of the coolingcurve of PET.

Sample	Δ H$_c$ [$\frac{J}{g}$]	T$_c$ [°C]
PET cc	-34.563	182.73

As can be seen in the attachment on pages 4 and 6, no crystallization enthalpy and no second melting enthalpy were observed. Thus, before the measurement both polymers are semi-crystalline, however, the chosen cooling conditions of 20 $\frac{°C}{min}$ are too fast for the PS to crystallize again and is therefore amorphous during the measurement of the second heating curve.

Using the melting enthalpy, the degree of crystallization α were calculated to

$$\alpha_{\text{PET}} = \frac{38{,}368\,\frac{\text{J}}{\text{g}}}{140\,\frac{\text{J}}{\text{g}}} = 0.274$$

$$\alpha_{it\text{-PS}} = \frac{24{,}194\,\frac{\text{J}}{\text{g}}}{87\,\frac{\text{J}}{\text{g}}} = 0.278$$

As noted previously, the *it*-PS could not crystallize under the cooling rate and therefore became amorphous. As a result, only a glass transition temperature and no melting temperature can be observed in the second heating curve.

2 Questions

2.1 Question 1

The "Big"number indicates how many monomers are needed to make one full 360° rotation. The small number in the index says how many 360° rotations are needed until the order gets repeated. For example in the case of 8_5 Polybutene. It needs 8 monomers to complete one 360° rotation and 5 full rotation until the order starts from the start.

2.2 Question 2

i-PP can be made by using the Ziegler-Natta-catalysis. In this case catalysts like $TiCl_3$ on $MgCl_2$-carrier are used in combination with aluminumalkenes. It can also be done via Metallocene-catalysis, this technique is specially used to control the tacticity and results in high graded i-PP. i-PS can also be synthesized using the Metallocene-catalysis. But can also be synthesized via Anionic-Polymerization.

2.3 Question 3

Commonly the degree of crystallization decides how stiff a polymer is. The higher the degree of crystallization, the harder and stiffer the polymer. It also gets less ductile and more dense because crystalline parts can be packed better than amorph parts.

2.4 Question 4

The Polymer is divided in different sections a back folding section(black), which would be the PS in our case and a amorph section(red), which in our case would be the PE sections.

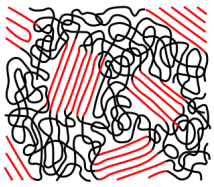


Abb. 1: The picture shows the part-.crystalline

2.5 Question 5

To get an semi-crystalline polymer out of an polymerglass it has to be reheated. By reheating the polymer the molecular mobility gets high enough to form crystals. This process is also known as cold crystallization.

2.6 Question 6

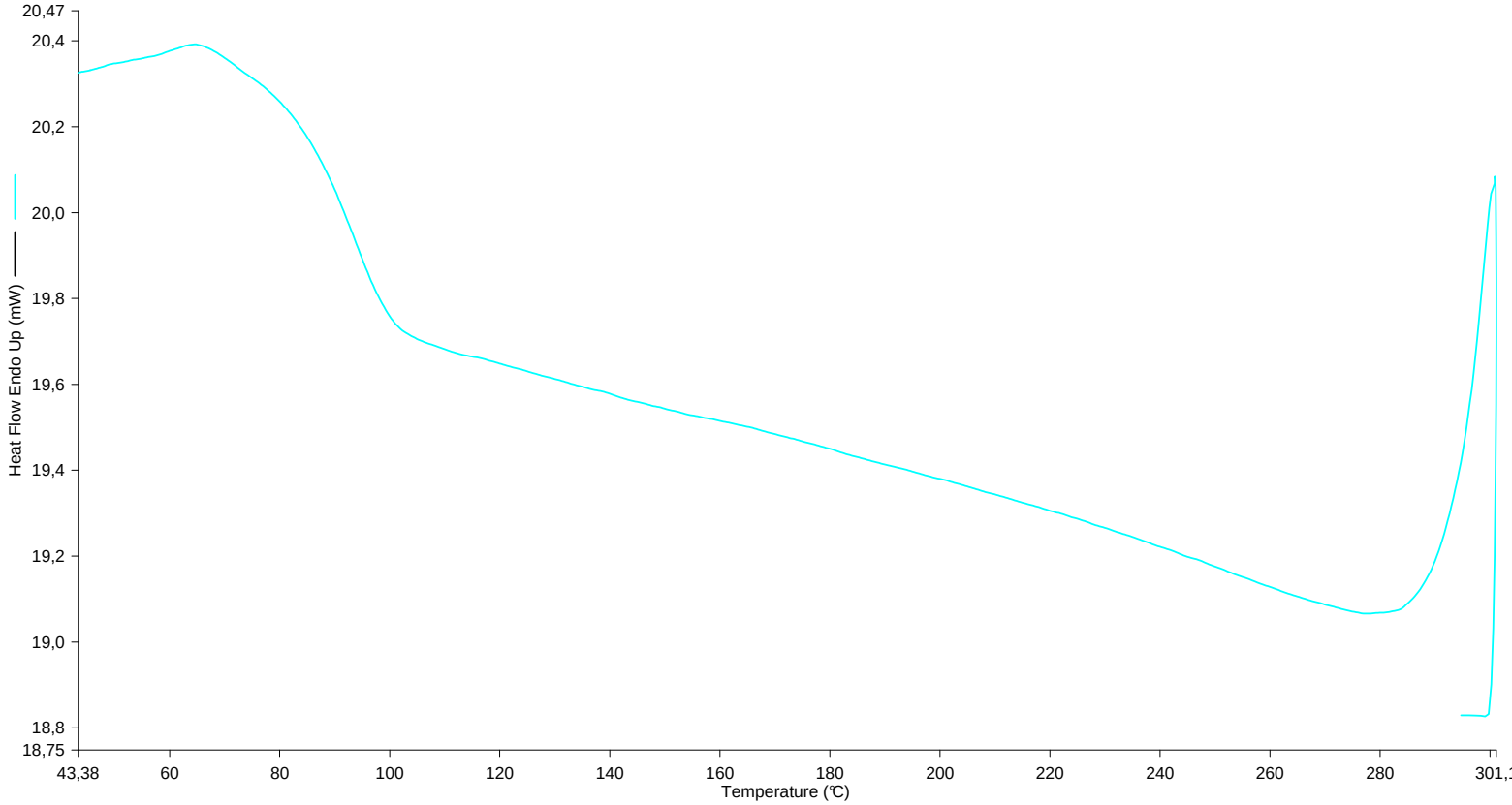
The increase in mass can be explained by oxidation. While increasing the temperature in the TGA the polymer oxidizes, the added Oxygen increases the mass of the polymer (molecule if the polymer gets destroyed).

2.7 Question 7

2.8 Question 8

Due to the residual moisture, the PET decomposes to Ethylenglycol and Terephthalic acid. The Ethylenglycol eliminates water which results in vinylalcohol, which can tautomerize to Ethanal.

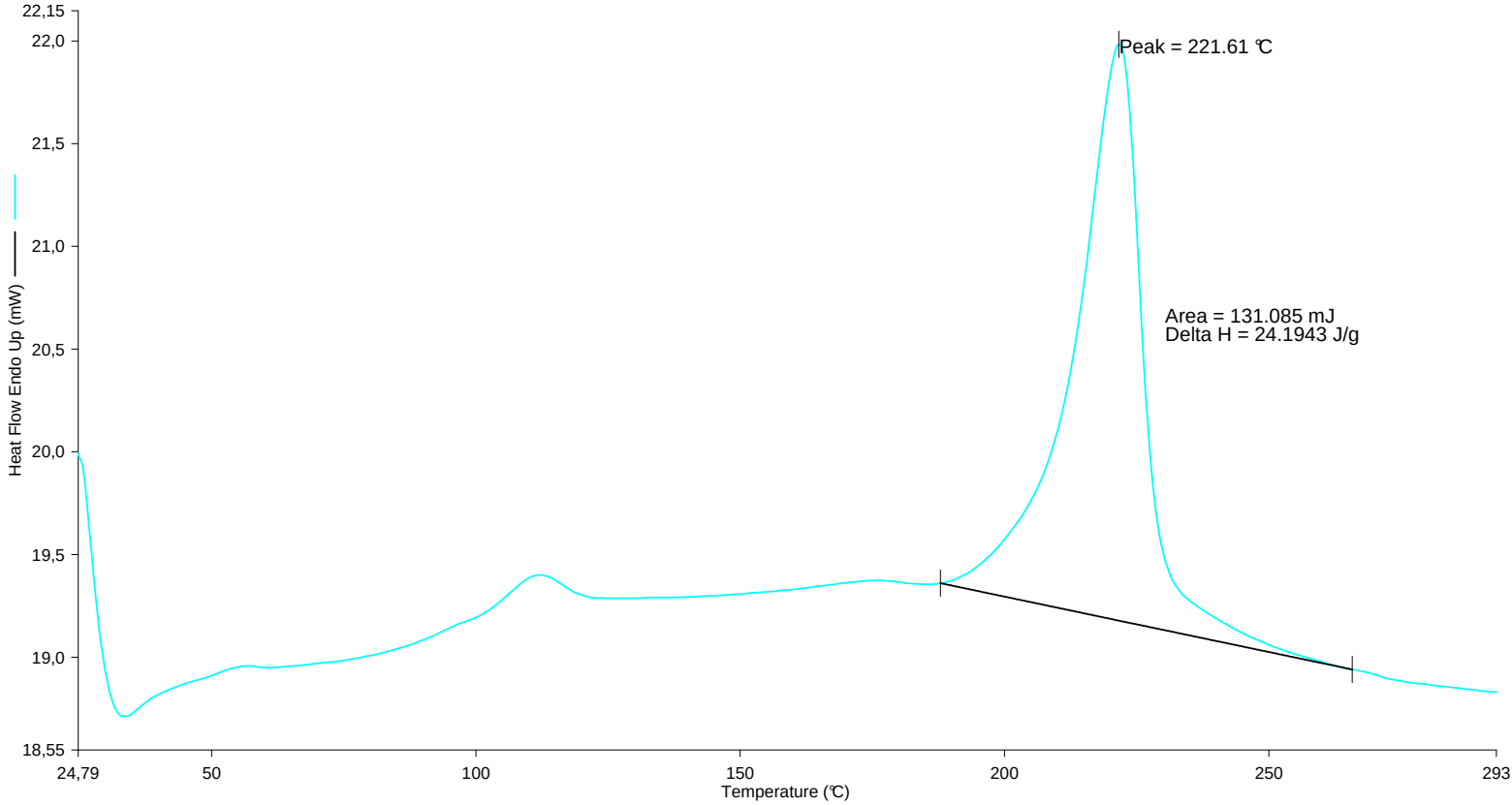
Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
Operator ID:	MD
Sample ID:	G8-ps-iso
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:26:12

1) Heat from 30.00°C to 300.00°C at 20.00°C/min
2) Cool from 300.00°C to 30.00°C at 20.00°C/min
3) Heat from 30.00°C to 300.00°C at 20.00°C/min

Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
Operator ID:	MD
Sample ID:	G8-ps-iso
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:25:16

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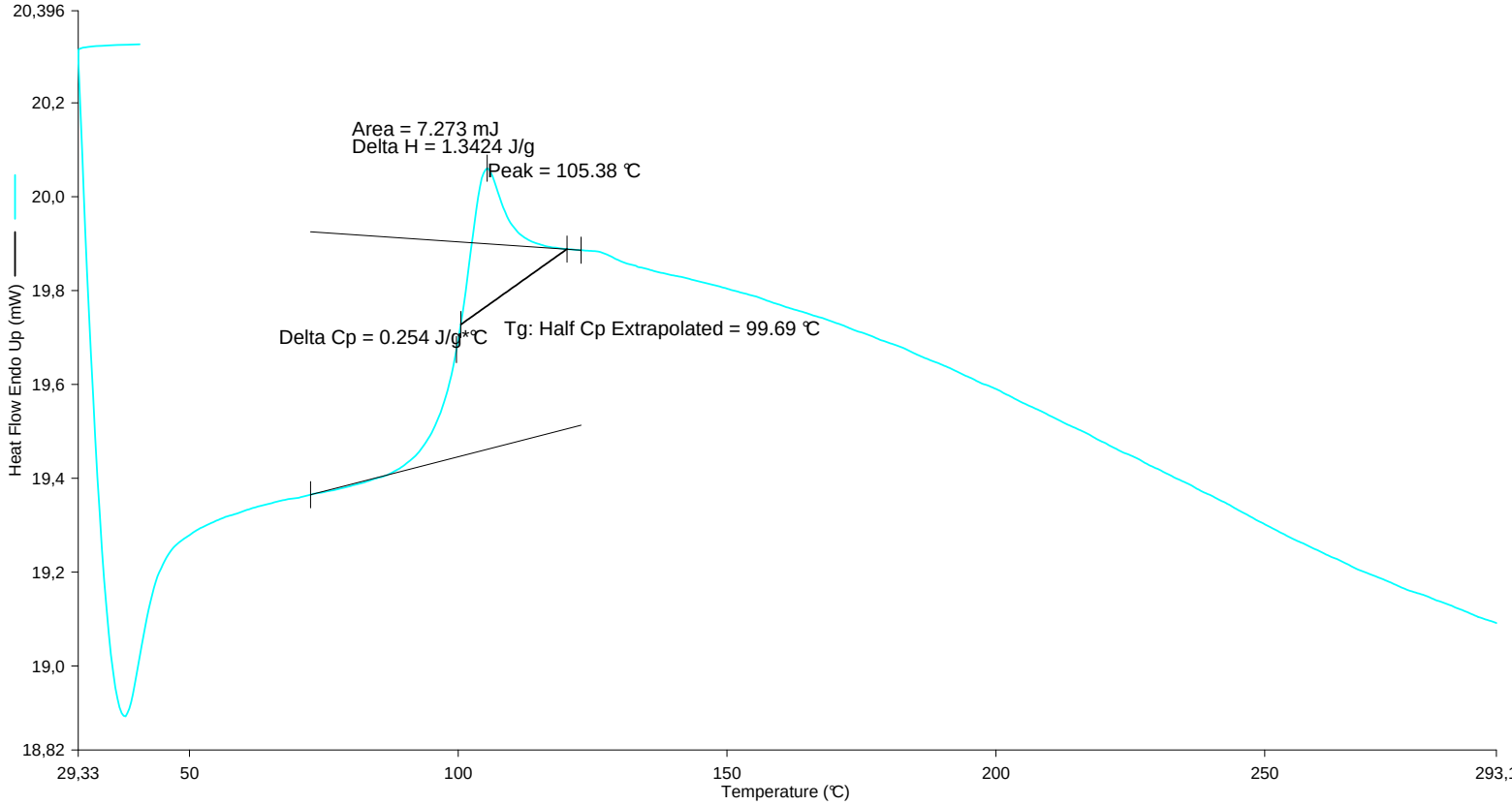
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G...\ps-iso-dsc.d6d
Operator ID:	MD
Sample ID:	G8-ps-iso
Sample Weight:	5.418 mg
Comment:	



10.03.2026 11:27:53

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3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
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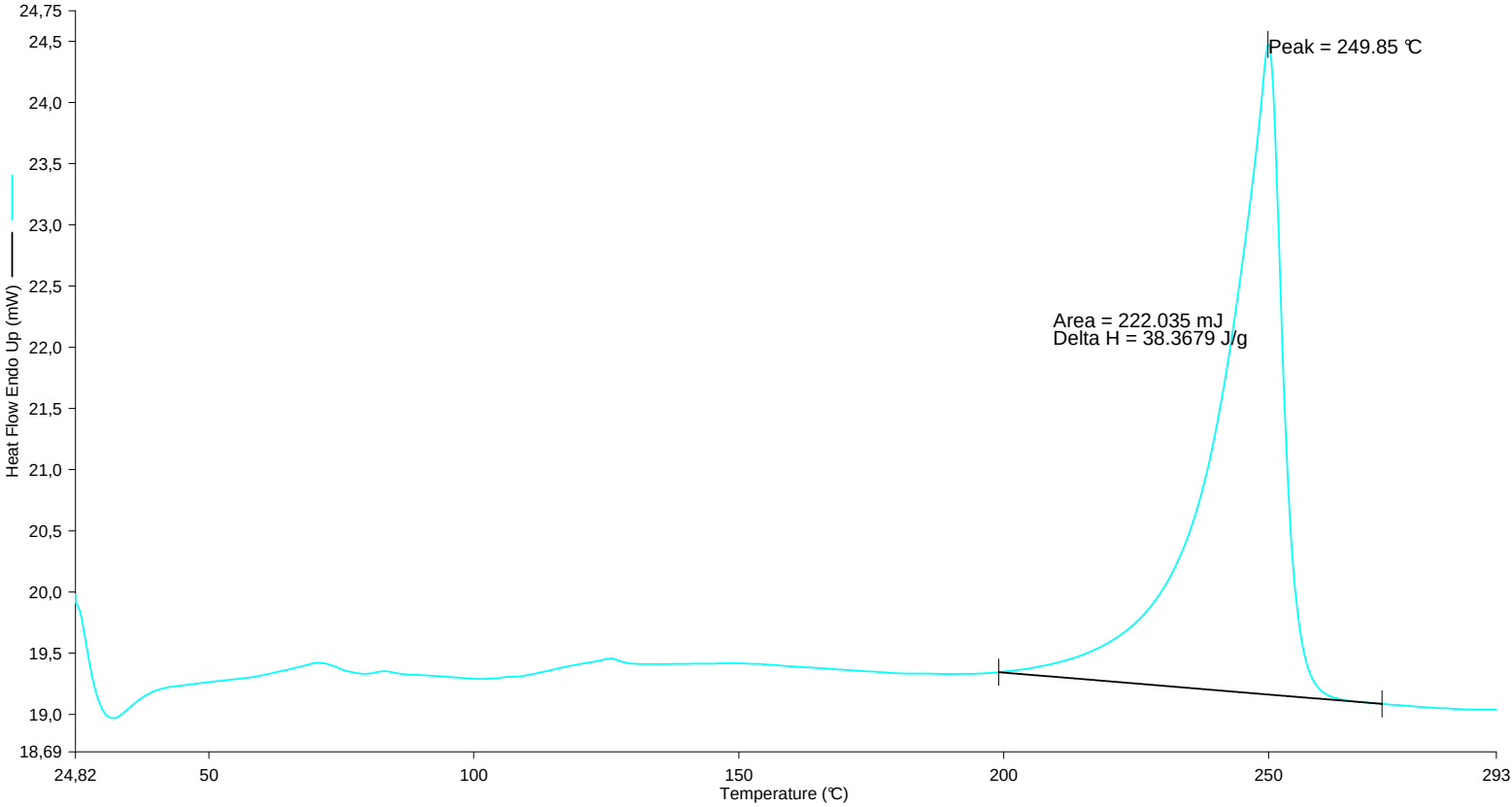
Sample ID:	G8-PET
Sample Weight:	5.787 mg
Comment:	



10.03.2026 10:26:23

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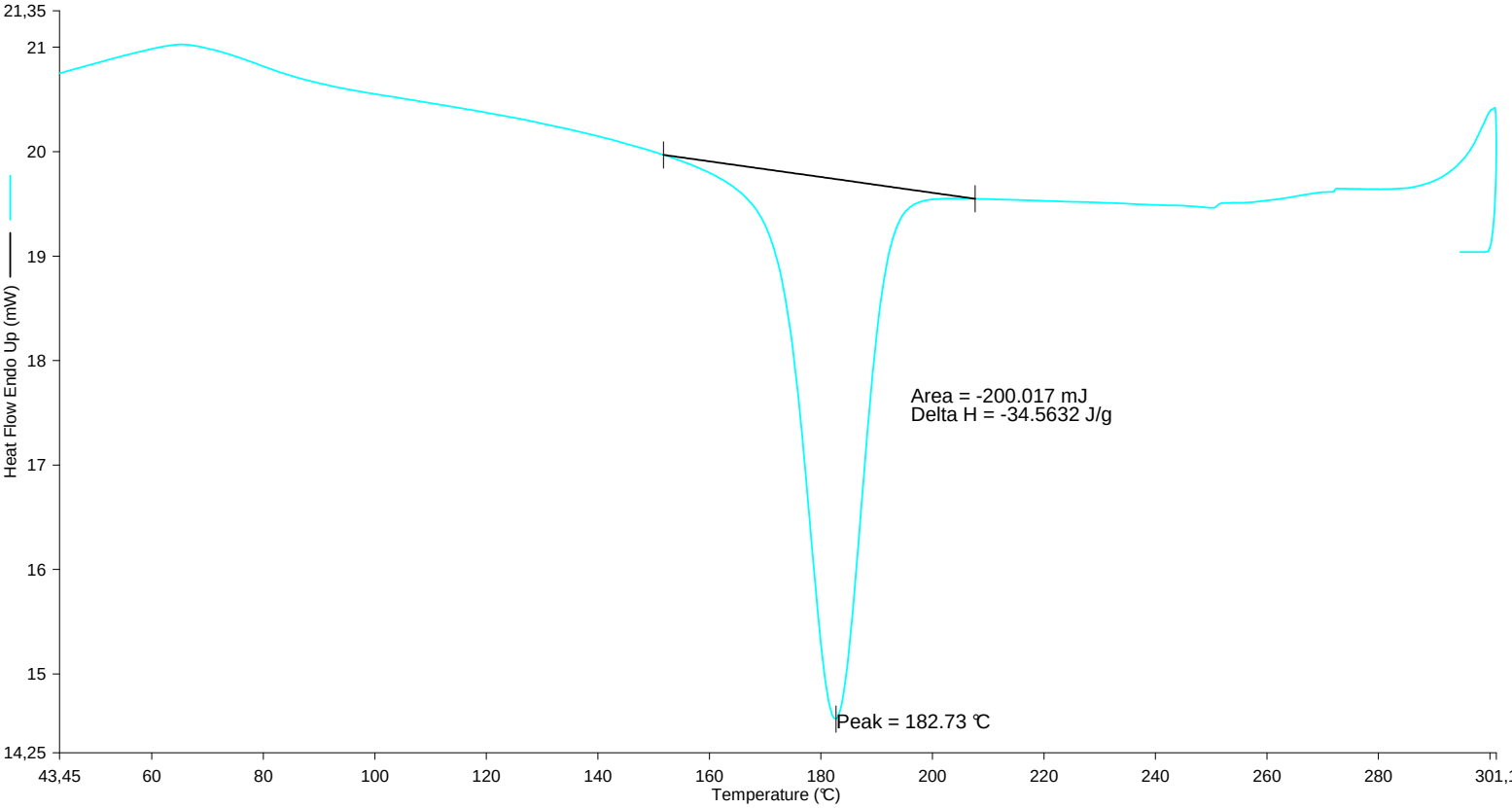
2) Cool from 300.00℃ to 30.00℃ at 20.00℃/min
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
Operator ID:	MD
Sample ID:	G8-PET
Sample Weight:	5.787 mg
Comment:	



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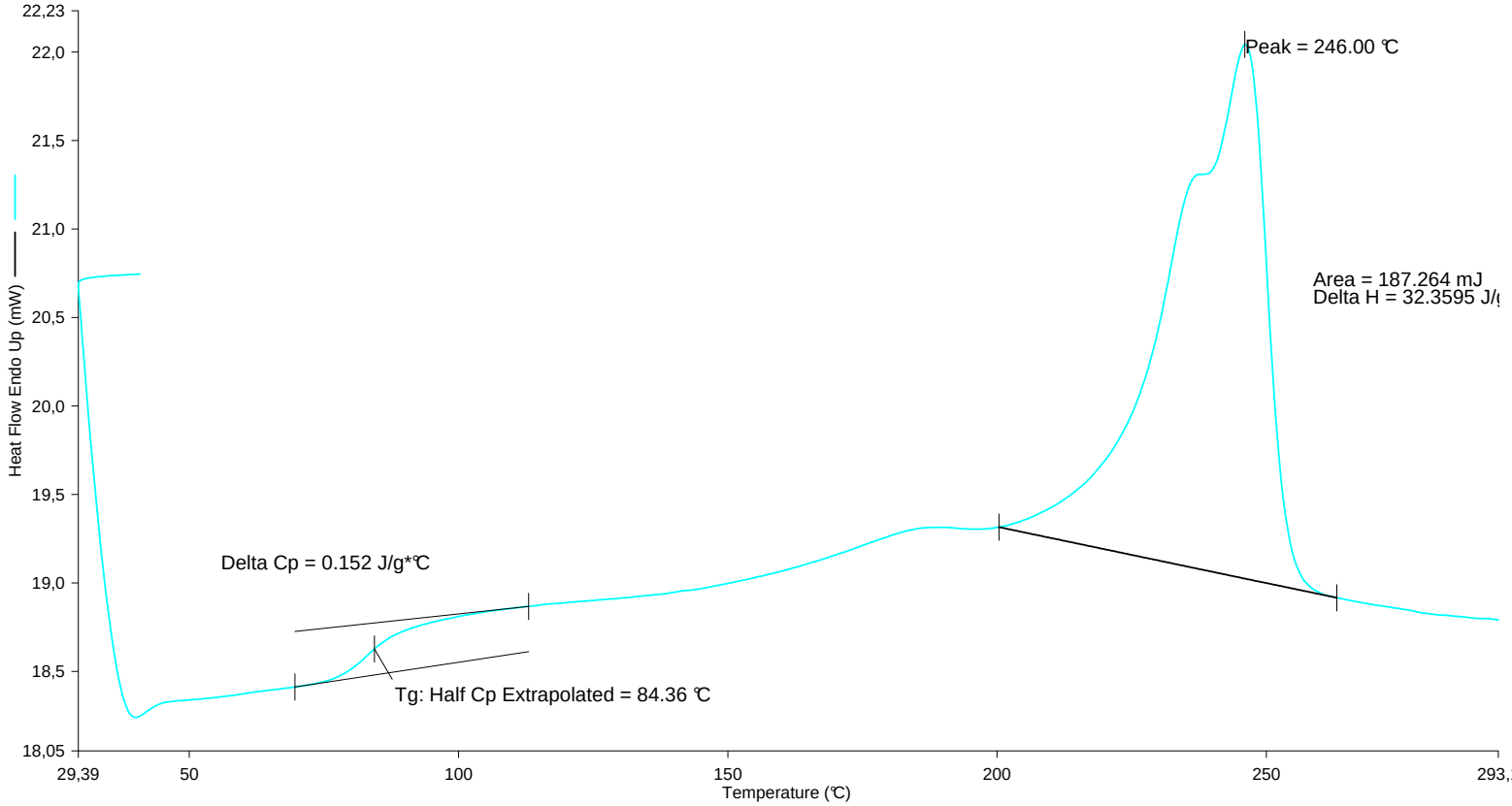
3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G8\pet-dsc.d6d
Operator ID:	MD
Sample ID:	G8-PET
Sample Weight:	5.787 mg
Comment:	



10.03.2026 10:29:27

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3) Heat from 30.00℃ to 300.00℃ at 20.00℃/min

Filename:	C:\Users\DSC\Desktop\MD\G...\G8-pet-tga.t6d
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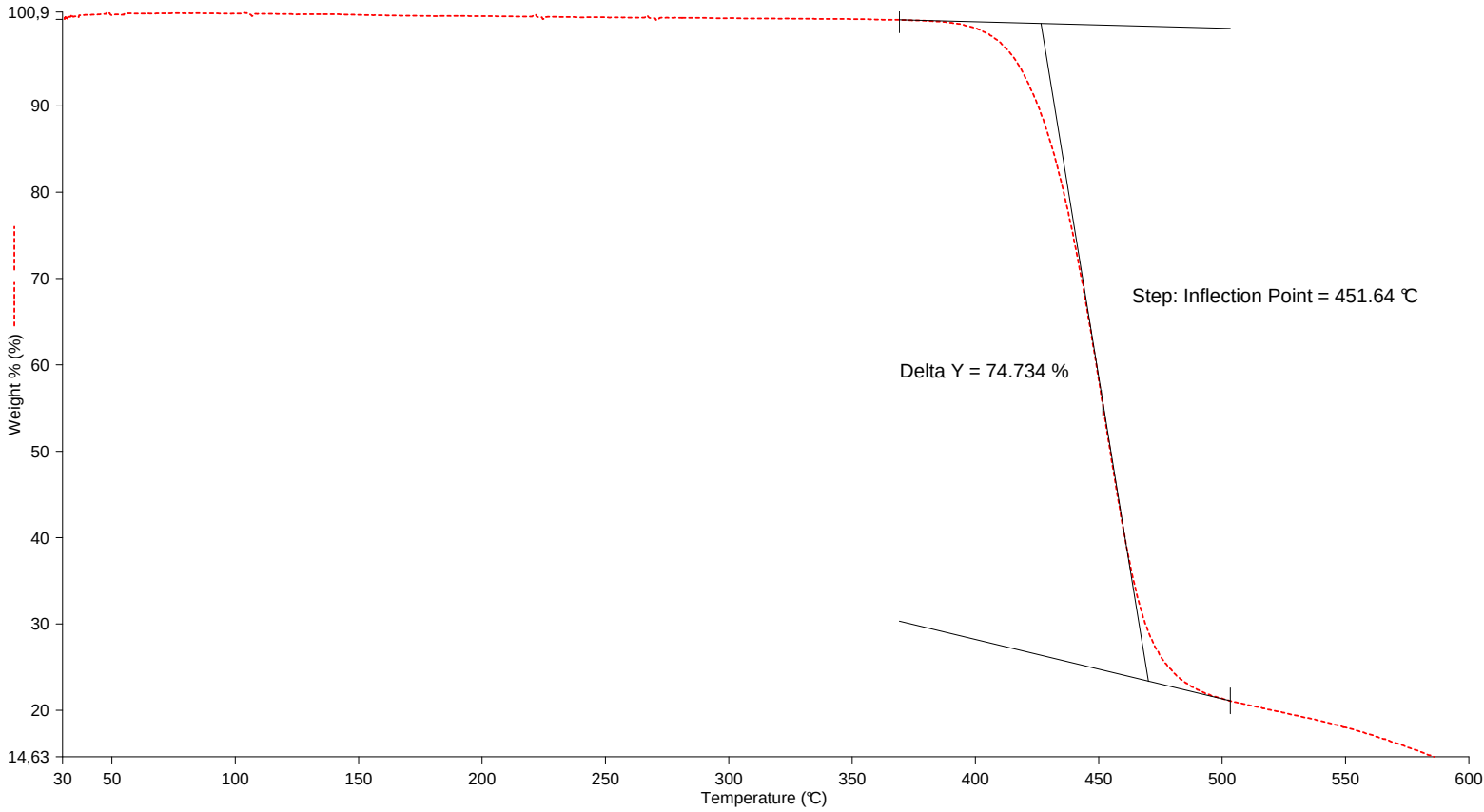
Sample ID:	G8-PET
Sample Weight:	3.597 mg
Comment:	



10.03.2026 10:42:27

1) Heat from 30.00℃ to 600.00℃ at 20.00℃/min